

GUNNARN, Sweden

WMO: 021260

Lat: 65.0094N

Lon: 17.6993E

Elev: 823

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-17.8	-28.0	1.2	-23.2	-22.1	1.7	-17.8	26.0	33.4	20.7	34.3	1.1	320	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	76.6	60.3	73.1	59.0	69.5	57.6	62.9	71.8	61.0	69.3	59.3	66.8	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.8	79.1	65.0	57.8	73.7	63.2	55.9	68.6	61.5	28.8	71.6	27.5	69.5	26.3	66.7	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	16.3	13.8	-29.7	80.8	5.0	3.6	-33.3	83.4	-36.3	85.6	-39.1	87.6	-42.7	90.3		
			-29.6	65.8	4.9	2.4	-33.1	67.5	-36.0	68.9	-38.7	70.2	-42.3	72.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	12.4	13.4	21.7	33.1	43.5	52.7	58.1	54.7	46.2	34.8	23.2	16.4
	DBStd	18.62	14.03	14.57	10.54	6.70	6.70	6.15	5.08	5.67	5.86	7.21	11.14	13.77
	HDD50	6318	1165	1025	878	508	224	38	1	18	142	472	805	1042
	HDD65	11221	1630	1445	1343	958	667	373	221	321	564	937	1255	1507
	CDD50	585	0	0	0	0	22	118	253	164	28	0	0	0
	CDD65	12	0	0	0	0	0	2	7	3	0	0	0	0
	CDH74	223	0	0	0	0	8	58	110	47	0	0	0	0
	CDH80	17	0	0	0	0	0	4	11	2	0	0	0	0
	WSAvg	5.1	4.3	4.5	5.7	5.7	6.0	6.2	5.2	4.6	5.0	4.8	4.1	4.6
	PrecAvg	23.5	1.7	1.3	1.3	1.1	1.6	2.8	3.8	3.3	1.5	2.0	1.8	1.4
Precipitation	PrecMax	34.0	3.5	3.3	2.1	2.9	3.0	4.6	6.5	6.8	2.7	3.4	4.0	3.1
	PrecMin	17.8	0.3	0.3	0.3	0.2	0.2	0.4	1.1	1.1	0.4	0.7	0.5	0.3
	PrecStd	5.0	0.7	0.9	0.6	0.7	0.8	1.3	1.6	2.2	0.7	0.9	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	43.0	46.3	60.4	73.9	80.2	81.4	78.7	67.2	54.1	48.2	42.6
		MCWB	38.3	38.4	38.8	46.5	56.3	60.3	63.1	62.6	56.8	48.5	45.3	39.3
	2%	DB	38.5	39.8	42.1	52.6	68.0	74.7	77.1	74.6	62.3	50.5	42.3	39.2
		MCWB	35.5	36.0	36.8	42.6	53.0	58.6	61.0	60.3	54.6	47.2	39.8	36.1
	5%	DB	34.9	35.9	39.6	48.2	62.5	69.8	73.7	70.2	58.8	48.5	39.1	36.5
		MCWB	32.7	32.8	35.0	40.0	50.5	55.9	59.6	59.2	52.7	45.6	37.0	34.3
	10%	DB	31.6	32.7	36.8	44.5	57.3	65.7	70.2	66.2	56.2	46.1	36.4	33.9
		MCWB	30.3	30.7	32.7	37.8	47.9	54.0	58.5	57.3	51.3	43.5	35.0	32.5
	0.4%	WB	38.5	38.5	39.8	47.0	57.9	63.3	66.3	64.7	59.1	50.2	44.9	39.6
		MCDB	42.0	42.3	44.8	58.1	69.5	75.7	74.1	72.3	64.4	52.2	47.6	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	36.3	37.4	43.3	55.0	60.0	63.5	62.2	56.1	48.1	40.3	36.5
		MCDB	38.3	39.7	41.7	50.9	65.2	70.6	72.1	70.6	60.1	50.3	42.1	38.6
	5%	WB	32.8	33.2	35.2	40.6	51.7	57.7	61.9	60.3	54.0	45.9	37.3	34.3
		MCDB	34.5	35.4	39.1	47.3	61.4	66.3	70.2	67.8	57.7	48.1	38.7	35.9
	10%	WB	30.2	30.9	33.1	38.3	48.4	55.4	59.9	58.6	52.2	43.6	35.0	32.5
		MCDB	31.5	32.7	36.3	43.6	55.8	63.1	67.4	65.3	55.4	45.7	36.2	33.8
Mean Daily Temperature Range	5% DB	MDBR	15.7	17.5	19.4	17.1	18.5	17.3	17.8	17.4	15.4	11.5	11.3	14.4
		MCDBR	16.0	15.1	18.4	24.5	29.3	24.8	24.6	24.7	19.9	14.5	10.9	15.0
		MCWBR	14.2	13.0	14.2	15.1	16.2	11.2	10.8	12.1	12.8	11.1	9.5	13.5
	5% WB	MCDBR	14.8	14.5	17.0	21.6	27.3	21.7	21.4	20.5	17.1	13.5	10.1	14.3
		MCWBR	13.4	12.8	13.4	13.7	15.8	10.7	10.6	11.2	11.9	10.9	9.0	13.0
Clear-Sky Solar Irradiance	taub		0.188	0.262	0.294	0.327	0.335	0.336	0.348	0.334	0.306	0.287	0.339	N/A
	taud		1.952	2.175	2.235	2.326	2.470	2.498	2.477	2.505	2.569	2.433	2.720	N/A
	Ebn at Noon		107	201	248	268	278	279	272	265	251	197	103	N/A
	Edh at Noon		7	17	26	31	30	30	29	26	20	14	6	N/A
All-Sky Solar Radiation	RadAvg		38	216	666	1219	1499	1657	1514	1135	643	258	61	12
	RadStd		3	35	62	88	167	128	169	120	82	41	11	1

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page